

Qi Zhang

Summary

Software Engineer with experience building backend services, AI-powered automation, and distributed systems. Experienced in developing reliable Python services, automating operational workflows, and improving developer productivity. Passionate about building scalable cloud infrastructure and AI platforms.

Professional Experience

Software Engineer | Microsoft(contract) | Redmond, WA | 12/2025 - 07/2026

- Windows Compatibility & Diagnostics Platform

- Designed and implemented Python/FastAPI backend services to automate Windows crash analysis workflows, processing crash dumps, execution traces, and diagnostic logs to reduce manual investigation effort.
- Built a modular data processing framework that collected and normalized diagnostic telemetry, extracting stack traces, runtime signals, and failure metadata to support scalable and consistent failure analysis across platform.
- Developed automated failure detection and classification workflows by correlating stack traces, loaded modules, and runtime diagnostics, improving debugging consistency and accelerating root cause investigation.
- Leveraged Azure OpenAI to automate failure triage and generate diagnostic hypotheses, reducing manual investigation effort by approximately ~ 50%.

Software Engineer | Flexport | Bellevue, WA | 10/2021 - 11/2023

- Work Management Platform

- Built the work management platform using Scala, Python and React to process logistics workflow
- Architected shipment process templates for forwarding application, establishing best practices that improved development efficiency by ~ 30%.
- Designed an event-driven processing system using Amazon SQS, improving throughput and reducing processing latency by ~ 20%.
- Developed gRPC APIs for inter-service communication, enabling reliable and low-latency messaging across distributed systems.
- Designed and implemented an automated document processing workflow that extracted customs information from prior work items, standardized processing logic, and introduced validation and proactive alerting, reducing manual processing time by ~ 25%.
- Integrated the Atlassian Jira API into the Work Management platform, enabling automated synchronization of engineering workflows and reducing manual operational effort by ~ 15%.

- Communication Platform

- Built the shipment communication platform using Python, Ruby on Rails, and React to streamline internal and external messaging workflows.
- Led frontend implementation of a notification settings system for individual and enterprise users using React and GraphQL.
- Monitored production services using Datadog dashboards, logs, and telemetry data to investigate workflow failures, validate system behavior, and improve operational reliability.
- Mentored and onboarded junior engineers, accelerating their ramp-up time and fostering a culture of continuous learning by leading regular technical brown-bag sessions on distributed systems.

Contacts

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Education

Northeastern University

- 01/2019 - 12/2020
- MS in Computer Science
- San Jose, CA

University of Washington

- 09/2014 - 03/2016
- MS in Mechanical Engineering
- Seattle, WA

Huazhong University of Science and Technology

- 09/2010 - 06/2014
- BS in Materials Engineering
- Wuhan, China

Skills & Abilities

Programming Languages

- C/C++
- C#
- Python
- Java
- Scala
- Ruby
- Go
- Kotlin
- HTML
- React

Frameworks & Tools

- Spring Boot
- Ruby on Rails
- React
- Node.js
- ZIO
- gRPC
- GraphQL
- Atlassian Jira API
- FastAPI